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Connector

Abstract

Provided is a connector with which waterproofness can be ensured. The connector includes: a housing main body **23** which includes terminal housing chambers **59** and an upper surface of which is closed off; terminal metal fittings **11** that are housed inside the terminal housing chambers **59**; a cover **15** that is locked to the housing main body **23** and arranged below the terminal housing chambers **59**; and eaves portions **62** that protrude laterally from an upper end of the housing main body **23.**

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based on and claims priority from Japanese Patent Application No. 2021-190339, filed on Nov. 24, 2021, with the Japan Patent Office, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a connector.

BACKGROUND

(3) A connector disclosed in JP H7-161411A includes a male connector housing and a female connector housing that can be fitted to one another. The male connector housing and the female connector housing each have terminal housing chambers formed therein. Note that techniques relating to connectors are also disclosed in JP H1-76123U, JP 2017-98019A, and JP 2020-80237A.

SUMMARY

(4) In the case of JP H7-16141 IA, in a state of use, the male connector housing is arranged on the upper side, and the female connector housing is arranged on the lower side. Furthermore, the terminal housing chambers of the two connector housings are arranged along the up-down direction. Rubber stoppers are housed inside the terminal housing chambers. However, if the rubber stoppers deteriorate or are damaged, water infiltrates into the terminal housing chambers and waterproofness cannot be ensured.

(5) In contrast, if the upper surface of the male connector housing is closed off, the infiltration of water into terminal housing chambers can be prevented. However, if a cover is arranged below the terminal housing chambers and a gap is formed between the cover and the male connector housing, there is a concern that water may infiltrate into the terminal housing chambers through the gap.
(6) In view of this, an object of the present disclosure is to provide a connector with which waterproofness can be ensured.
(7) A connector according to the present disclosure includes: a housing main body which includes a terminal housing chamber and an upper surface of which is closed off; a terminal metal fitting that is housed inside the terminal housing chamber, a cover that is locked to the housing main body and arranged below the terminal housing chamber, and an eaves portion that protrudes laterally from an upper end of the housing main body.
(8) According to the present disclosure, a connector with which waterproofness can be ensured can be provided.
(9) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a rear view of a connector fitted to a mating connector in embodiment 1 of the present disclosure.
(2) FIG. **2** is a lateral view of the connector fitted to the mating connector.
(3) FIG. **3** is a cross-sectional view taken along line X-X in FIG. **2**.
(4) FIG. **4** is a cross-sectional view taken along line Y-Y in FIG. **2**.
(5) FIG. **5** is a cross-sectional view taken along line Z-Z in FIG. **2**.
(6) FIG. **6** is a lateral cross-sectional view of the connector.
(7) FIG. **7** is a perspective view of a terminal metal fitting connected to an end portion of an electric wire.
(8) FIG. **8** is a rear view of a housing.
(9) FIG. **9** is a bottom view of the housing.
(10) FIG. **10** is a front view of the housing.
(11) FIG. **11** is a perspective view of an upper inner member.
(12) FIG. **12** is a perspective view of a lower inner member.
(13) FIG. **13** is a perspective view of a cover.
DETAILED DESCRIPTION
(14) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. The illustrative embodiments described in the detailed description, drawings, and claims are not meant to be limiting. Other embodiments may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented here.
(15) First, aspects of the present disclosure will be listed and described.
(16) (1) A connector according to the present disclosure includes: a housing main body which includes a terminal housing chamber and an upper surface of which is closed off; a terminal metal fitting that is housed inside the terminal housing chamber; a cover that is locked to the housing main body and arranged below the terminal housing chamber, and an eaves portion that protrudes laterally from an upper end of the housing main body.
(17) According to the above-described configuration, the infiltration of water into the terminal housing chamber through a gap between the cover and the housing main body can be prevented

because the upper surface of the housing main body is closed off and the eaves portion protrudes laterally from the upper end of the housing main body.
(18) (2) Preferably, the eaves portion is shaped so as to protrude laterally further than a locked portion where the housing main body and the cover are locked to one another does, above the locked portion.
(19) Although a gap is formed in the locked portion where the housing main body and the cover are locked to one another, the infiltration of water into the gap can be avoided because the eaves portion protrudes laterally above the locked portion according to the above-described configuration.
(20) (3) Preferably, an upper surface of the eaves portion is continuous with the upper surface of the housing main body without any level differences.
(21) According to the above-described configuration, water falling on the upper surface of the housing main body is allowed to move smoothly onto the upper surface of the eaves portion.
(22) (4) Preferably, a hanging-down portion that protrudes downward from a laterally-protruding end side of the eaves portion is included.
(23) According to this configuration, the infiltration of water into the terminal housing chamber of the housing main body can be more reliably prevented because water is allowed to fall down along the hanging-down portion from the upper surface of the eaves portion.
(24) (5) Preferably, the housing main body includes a pair of locking walls, and a cover housing portion an upper surface of which is closed off is formed between the pair of locking walls in the housing main body, and the cover is arranged inside the cover housing portion, and includes lateral walls that face inner surfaces of the locking walls and that are locked to the locking walls.
(25) Although gaps are formed between the lateral walls and the locking walls, the infiltration of water into the gaps can be avoided because the lateral walls of the cover are arranged so as to face the inner surfaces of the locking walls of the housing main body inside the cover housing portion according to the above-described configuration.
(26) Preferably, the connector is a connector that is provided to a roof panel of a vehicle, and the eaves portion is arranged along a lower surface of the roof panel.
(27) According to the above-described configuration, the infiltration of water into the terminal housing chamber of the housing main body can be prevented because dew on the roof panel is allowed to freely fall directly from the eaves portion.
DETAILS OF EMBODIMENTS OF PRESENT DISCLOSURE
(28) Specific examples of embodiment of the present disclosure will be described below with reference to the drawings. Note that the present invention is not limited to these examples, and is intended to include all modifications that are indicated by the claims and are within the meaning and scope of equivalents of the claims.
Embodiment 1
(29) As illustrated in FIGS. **1** and **2**, a connector **10** according to embodiment 1 of the present disclosure is arranged below a roof panel **90** of a vehicle such as an automobile, which is not illustrated in entirety. In other words, the connector **10** is arranged inside the vehicle. A mating connector **8** that is connected to a circuit board **100** is arranged above the roof panel **90**, or in other words, outside the vehicle. As illustrated in FIGS. **2** and **6**, the connector **10** and the mating connector **8** are fitted to one another through an opening **91** having a rectangular cross-sectional shape that is formed in the roof panel **90**. The connector **10** includes terminal metal fittings **11**, a housing **12**, an upper inner member **13**, a lower inner member **14**, and a cover **15**. The terminal metal fittings **11** are connected to terminal portions of electric wires **200**. The electric wires **200** are routed along the lower surface of the roof panel **90**.
(30) In the following description, note that, in regard to the front-rear direction, the side in which the terminal metal fittings **11** are connected to the electric wires **200** is the front side. In FIGS. **2**, **6**, and **7**, and FIGS. **9** to **13**, the reference symbol F is the front side. The reference symbol R is the

rear side. In regard to the up-down direction, the side in which the mating connector **8** is located relative to the connector **10** is the upper side. In FIGS. **1**, **2**, **6**, and **7**, and FIGS. **10** to **13**, the reference symbol U is the upper side. The reference symbol D is the lower side. The up-down direction corresponds to the vertical direction (direction of gravity), and matches the up-down direction in a state of use in present embodiment 1.
(31) Terminal Metal Fittings
(32) Each terminal metal fitting **11** is a shield terminal, and has an L-shape in a lateral view, as illustrated in FIG. **7**. The terminal metal fitting **11** includes an unillustrated electroconductive inner conductor, an electroconductive outer conductor **16**, and an insulating dielectric member **83** (see FIG. **7**) that is arranged between the inner conductor and the outer conductor **16**. The inner conductor and the outer conductor **16** are electrically insulated from one another with the dielectric member **83** therebetween. The outer conductor **16** includes a cylinder portion **17** whose axis is oriented in the up-down direction, and an extension portion **18** that intersects the cylinder portion **17** and extends toward the rear. In the rear part of the extension portion **18**, a sheath barrel **21** and a shield barrel **19** forming an open barrel shape are formed so as to be arranged one in front of the other. Although not described in detail, the terminal metal fittings **11** consist of large-diameter shield terminals including one shield barrel **19** (the terminal metal fittings **11** in the middle and upper tiers in FIG. **6**) and small-diameter shield terminals including two shield barrels **19** (the terminal metal fitting **11** in the lower tier in FIG. **6**).
(33) Each electric wire **200** is a coaxial cable, and, as illustrated in FIG. **1**, includes an electroconductive core wire a, an insulating coating b that covers the outer circumference of the core wire a, an electroconductive shield portion c made from a braided wire or the like that covers the outer circumference of the coating b, and an insulating sheath d that covers the outer circumference of the shield portion c. In the terminal portion of the electric wire **200**, the shield portion c is exposed as a result of the sheath d being removed, and furthermore, the core wire a is exposed as a result of the coating b being removed.
(34) As illustrated in FIG. **7**, the shield barrel **19** is crimped and electrically connected to the shield portion c of an electric wire **200** from over a sleeve **85**. The sheath barrel **21** is crimped and mechanically connected to the sheath d of the electric wire **200**. In the upper wall portion in the front part of the extension portion **18**, a protruding piece **22** is formed by being cut out and raised.
(35) Although not illustrated, the rear part of the inner conductor is crimped and electrically connected to the core wire a of the electric wire **200**. Furthermore, the front part of the inner conductor is arranged along the up-down direction inside the cylinder portion **17** together with the front part of the dielectric member **83**, and is electrically connected to a mating inner conductor of a mating terminal metal fitting **95** (see FIG. **6**) that is attached to the mating connector **8**.
(36) Note that the connector **10** also includes an unillustrated general terminal that does not have a shield function and that is entirely made from an electroconductive metal plate. The general terminal is connected to a coated electric wire **250** (see FIG. **1**) that does not include the shield portion c, and is also connected to an unillustrated tab that is attached to the mating connector **8**.
(37) Housing
(38) The housing **12** is made from synthetic resin, and has an L-shape in a lateral view, as illustrated in FIG. **6**. The housing **12** includes a housing main body **23** in the rear part thereof, and includes a fitting portion **24** in the front part thereof. The height of the housing main body **23** is lower than the height of the fitting portion **24**. As illustrated in FIG. **9**, the width of the housing main body **23** is larger than the width of the fitting portion **24**.
(39) As illustrated in FIG. **6**, the fitting portion **24** is fitted into guide walls **9** formed in the mating connector **8**. As illustrated in FIGS. **9** and **10**, a lock arm **25** that can deform by bending is formed on the front surface of the fitting portion **24**. The lock arm **25** is formed so as to extend downward from the upper end portion of the fitting portion **24**. The lock arm **25** can lock the mating connector **8**. A pair of front-side locking portions **26** are formed on the lower end portions of the left and right

outer lateral surfaces of the fitting portion **24**. As illustrated in FIG. **2**, front-side locked portions **27** of the cover **15** can be locked to the front-side locking portions **26**.

(40) Although not illustrated in detail, a staircase portion **28** (see FIGS. **6** and **9**) that has the shape of a staircase in a lateral cross-sectional view is formed inside the fitting portion **24**. The staircase portion **28** includes an upper-tier surface **29**, a middle-tier surface **31**, and a lower-tier surface **32** that become gradually lower in height from the rear toward the front. Furthermore, a plurality of terminal insertion holes **33** having circular cross-sectional shapes that extend in the up-down direction are formed inside the fitting portion **24**. As illustrated in FIG. **6**, the cylinder portion **17** of a terminal metal fitting **11** is inserted from below into each terminal insertion hole **33**.

(41) The rear-side terminal insertion holes **33** open to the upper-tier surface **29** of the staircase portion **28**. The terminal insertion holes **33** between the front and rear sides open to the middle-tier surface **31** of the staircase portion **28**. The front-side terminal insertion holes **33** open to the lower-tier surface **32** of the staircase portion **28**.

(42) As illustrated in FIG. **9**, in the rear part of each of the upper-tier surface **29**, the middle-tier surface **31**, and the lower-tier surface **32** of the staircase portion **28**, recesses **34** are formed at positions corresponding to the terminal insertion holes **33**. In a surface that is continuous with the lower surface of a later-described upper wall **37** above the upper-tier surface **29**, a pair of left and right protruding-piece receiving portions **35** that are open in the shape of slits are formed. As illustrated in FIG. **6**, protruding pieces **22** of terminal metal fittings **11** are fitted into the protruding-piece receiving portions **35** and positioned relative to the housing **12**. Note that portions into which protruding pieces **22** are fitted similarly to the protruding-piece receiving portions **35** are also formed in the upper inner member **13** and the lower inner member **14** (see FIG. **6**).

(43) As illustrated in FIGS. **3** and **8**, the housing main body **23** includes a pair of locking walls **36** that face one another in the left-right direction, and the upper wall **37**, which is provided so as to extend across the upper ends of the locking walls **36**. The left and right outer lateral surfaces of the locking walls **36** are arranged along the up-down direction. A cover housing portion **38** is formed between the locking walls **36** and below the upper wall **37** in the housing main body **23**. As illustrated in FIG. **9**, the upper surface of the cover housing portion **38** is closed off by the upper wall **37**, and the lower and rear sides of the cover housing portion **38** are open. Furthermore, the cover housing portion **38** is formed so as to extend across the housing main body **23** and the fitting portion **24**, and the front side of the cover housing portion **38** communicates with the terminal insertion holes **33** in the fitting portion **24**.

(44) As illustrated in FIG. **9**, the locking walls **36** include front surface portions **39**, rear surface portions **41**, top surface portions **42**, and lateral surface portions **43** that define the rear part of the cover housing portion **38**. The front surface portions **39** orthogonally intersect and continue from the inner surfaces of the left and right lateral wall portions of the fitting portion **24**. The rear surface portions **41** are arranged in the rear end side of the housing **12** so as to face the front surface portions **39**. The top surface portions **42** each continue from the upper ends of a front surface portion **39** and a rear surface portion **41**, and are arranged so as to be flat along the left-right direction as illustrated in FIG. **3**.

(45) As illustrated in FIG. **3**, in the left and right inner lateral surfaces (inner surfaces) of the locking walls **36**, the lateral surface portions **43** include upper portions **44** that are arranged on the upper side, lower portions **45** that are arranged on the lower side, and step portions **46** that are each arranged between an upper portion **44** and a lower portion **45**. The upper portions **44** and the lower portions **45** are arranged along the up-down direction. The step portions **46** are arranged along the left-right direction. The lower portions **45** are located further outward toward the left and right than the upper portions **44** are as a result of the step portions **46** being interposed.

(46) As illustrated in FIG. **5**, lower inner lock portions **47** are formed in the lower portions **45** of the lateral surface portions **43** of the locking walls **36**. As illustrated in FIG. **5**, the lower inner lock portions **47** have the shape of tabs, protrude laterally from the main body portions of the lower

portions **45**, and are arranged inside the cover housing portion **38**. Lower-tier extraction holes **48** are formed in the locking walls **36** at positions corresponding to the front sides of the lower portions **45** (see FIG. **2**). The lower-tier extraction holes **48** are formed as a result of unillustrated molds for forming the upper surfaces of the lower inner lock portions **47** being extracted.

(47) As illustrated in FIG. **2**, locking portions **51** are formed in the locking walls **36** rearward of the lower inner lock portions **47**. The locking portion **51** have the shape of rectangular frames, and are formed between a pair of front and rear slits **81**, which are formed so as to extend upward from the lower ends of the locking walls **36**, so that the locking portions **51** can deform by bending in the left-right direction. In the center of each locking portion **51**, a locking hole **52** having a rectangular cross-sectional shape is formed so as to extend through the locking portion **51**. The inner ends of the locking holes **52** open to the lower portions **45**. As illustrated in FIG. **3**, later-described locked portions **53** of the cover **15** can be fitted into the locking holes **52** in the locking portions **51**.

(48) As illustrated in FIG. **2**, upper inner lock portions **54** are formed in the upper portions **44** of the lateral surface portions **43** of the locking walls **36**. As illustrated in FIG. **4**, the upper inner lock portions **54** have the shape of tabs, protrude laterally from the main body portions of the upper portions **44**, and are arranged inside the cover housing portion **38**. Upper-tier extraction holes **55** are formed in the locking walls **36** at positions corresponding to between the front and rear sides of the upper portions **44**. The upper-tier extraction holes **55** are formed as a result of unillustrated molds for forming the upper surfaces of the upper inner lock portions **54** being extracted.

(49) The upper surface of the upper wall **37** is formed so as to be flat along the front-rear direction and the left-right direction. As illustrated in FIG. **9**, a partition wall **57** is formed so as to protrude from a middle portion between the left and right sides of the lower surface of the upper wall **37**. The partition wall **57** is formed so as to extend in the front-rear direction from the housing main body **23** to the fitting portion **24**. A pair of left and right terminal housing chambers **59** are formed between the partition wall **57** and the lateral surface portions **43** below the upper wall **37**. As illustrated in FIG. **6**, the lower sides of the terminal housing chambers **59** are closed off by a later-described upper base portion **61** of the upper inner member **13**. The extension portions **18** of the terminal metal fittings **11** in the upper tier are arranged in the terminal housing chambers **59**.

(50) Furthermore, as illustrated in FIG. **8**, the housing main body **23** includes a pair of eaves portions **62** that protrude from the upper end of the housing main body **23** toward the left and right outer sides (laterally). The eaves portions **62** are long in the front-rear direction, and are formed to have rectangular shapes in a plan view. The upper surfaces of the eaves portions **62** are arranged along the front-rear direction and the left-right direction, and are continuous with the upper surface of the upper wall **37** without any level differences. The left and right outer protruding ends (outer ends) of the eaves portions **62** form the outermost ends of the housing **12** in the left-right direction. In other words, the eaves portions **62** are shaped so as to protrude laterally further than the left and right outer lateral surfaces of the locking walls **36** do, above the locking walls **36**.

(51) Furthermore, as illustrated in FIG. **8**, the housing main body **23** includes a pair of hanging-down portions **86** that protrude downward from the protruding end sides of the eaves portions **62**. The left and right surfaces of the hanging-down portions **86** are arranged along the up-down direction. As illustrated in FIG. **2**, the hanging-down portions **86** have the shape of rectangular plates that are long in the front-rear direction in a lateral view, and are formed over the entire front-rear-direction length of the eaves portions **62**. The length in which the hanging-down portions **86** protrude downward is set so as to be greater than the length in which the eaves portions **62** protrude laterally.

(52) Upper Inner Member

(53) The upper inner member **13** is made from synthetic resin, and, as illustrated in FIG. **11**, includes the upper base portion **61** having the shape of a flat plate, a pair of upper lateral walls **63** protruding downward from the left and right outer ends of the upper base portion **61**, and an upper partition wall **64** that protrudes downward from the center of the upper base portion **61** between the

left and right sides. As illustrated in FIG. **6**, the upper inner member **13** is attached to the inside of the housing **12**, and is arranged between the terminal metal fittings **11** in the upper tier and the terminal metal fittings **11** in the middle tier. A pair of left and right terminal housing chambers **59** are formed between the upper partition wall **64** and the upper lateral walls **63** below the upper base portion **61**. As illustrated in FIG. **6**, the lower sides of the terminal housing chambers **59** are closed off by a later-described lower base portion **65** of the lower inner member **14**. The extension portions **18** of the terminal metal fittings **11** in the middle tier are arranged in the terminal housing chambers **59**.
(54) As illustrated in FIG. **11**, a rib **66** that extends along the front-rear direction is formed in the center of the upper surface of the upper base portion **61** between the left and right sides. As illustrated in FIG. **1**, the rib **66** of the upper base portion **61** is arranged so as to be capable of coming into contact with the lower end of the partition wall **57** along the front-rear direction. As illustrated in FIG. **6**, the front end of the upper base portion **61** is arranged so as to be capable of coming into contact with a middle-tier wall surface **67** of the staircase portion **28** inside the fitting portion **24**.
(55) As illustrated in FIG. **11**, upper guide portions **68** having the shape of slopes are formed by chamfering the left and right outer ends of the upper surfaces of the upper lateral walls **63**. Furthermore, upper inner lock receiving portions **69** are formed below the upper guide portions **68** of the upper lateral walls **63**. The upper inner lock receiving portions **69** have the shape of quadrilateral recesses, and are open toward the lower side of the upper lateral walls **63**. As illustrated in FIG. **4**, the upper inner lock receiving portions **69** can be locked to the upper inner lock portions **54**.
(56) Lower Inner Member
(57) The lower inner member **14** is made from synthetic resin, and, as illustrated in FIG. **12**, includes the lower base portion **65** having the shape of a flat plate, a pair of lower lateral walls **71** protruding downward from the left and right outer ends of the lower base portion **65**, and a pair of left and right lower partition walls **72** that protrude downward from the center of the lower base portion **65** between the left and right sides. As illustrated in FIG. **6**, the lower inner member **14** is attached to the inside of the housing **12**, and is arranged between the terminal metal fittings **11** in the middle tier and the terminal metal fittings **11** in the lower tier. As illustrated in FIG. **12**, three terminal housing chambers **59** arranged in the left-right direction are formed below the lower base portion **65**, between the lower partition walls **72** and the lower lateral walls **71** and between the adjacent lower partition walls **72**. As illustrated in FIG. **6**, the lower sides of the terminal housing chambers **59** are closed off by a later-described cover wall **73** of the cover **15**. The extension portions **18** of the terminal metal fittings **11** in the lower tier and the coated electric wire **250** (see FIG. **1** and FIGS. **3** to **5**) connected to the general terminal are housed in the terminal housing chambers **59**.
(58) The upper surface of the lower base portion **65** is arranged so as to be capable of coming into contact with the lower end of the upper partition wall **64** along the front-rear direction. As illustrated in FIG. **6**, the front end of the lower base portion **65** is arranged so as to be capable of coming into contact with a lower-tier wall surface **74** of the staircase portion **28** inside the fitting portion **24**.
(59) As illustrated in FIG. **12**, a pair of left and right protruding portions **75** are formed so as to protrude from the lower lateral walls **71**. The protruding portions **75** have the shape of quadrilateral blocks. Lower guide portions **76** having the shape of slopes are formed by chamfering the left and right outer ends of the upper surfaces of the protruding portions **75**. Lower inner lock receiving portions **77** are formed below the lower guide portions **76** of the protruding portions **75**. The lower inner lock receiving portions **77** have the shape of quadrilateral recesses, and are open toward the lower side of the lower lateral walls **71**. As illustrated in FIG. **5**, the lower inner lock receiving portions **77** can be locked to the lower inner lock portions **47**.

(60) Cover
(61) The cover **15** is made from synthetic resin, and, as illustrated in FIG. **13**, includes the cover wall **73** having the shape of a flat plate. The up-down-direction thickness (plate thickness) of the cover wall **73** is thinner than the up-down-direction thicknesses of each of the upper base portion **61** and the lower base portion **65**. As illustrated in FIGS. **3** to **6**, the cover wall **73** is fitted inside the housing main body **23** and the fitting portion **24** so as to cover the lower-surface openings in the housing main body **23** and the fitting portion **24**. As illustrated in FIG. **5**, the lower side of the cover housing portion **38** is closed off by the cover wall **73**. As illustrated in FIG. **13**, the rear side of the cover wall **73** includes a pair of left and right protruding pieces **78** that protrude laterally further than the front side of the cover wall **73** does. The protruding pieces **78** are arranged inside an inner space that is defined by the front surface portions **39** of the locking walls **36**, and the rear surface portions **41**, the top surface portions **42**, and the lateral surface portions **43** in the cover housing portion **38** (see FIG. **3**; see FIG. **9** in regard to the front surface portions **39** and the rear surface portions **41**).
(62) A pair of left and right lateral walls **79** protruding upward from the left and right outer ends of the protruding pieces **78** are formed in the cover wall **73**. The lateral walls **79** have the shape of small plates extending along the front-rear direction and the up-down direction, and are arranged inside the housing main body **23** rearward of the protruding portions **75** of the lower inner member **14** and between the locking walls **36** and the lower lateral walls **71** in the cover housing portion **38** as illustrated in FIG. **3**. Gaps between the lateral walls **79** and the locking walls **36** are smaller than gaps between the lateral walls **79** and the lower lateral walls **71**. A pair of left and right locked portions **53** are formed so as to protrude from the outer surfaces of the lateral walls **79**. The locked portions **53** have the shape of tabs that can fit into the locking holes **52** in the locking portions **51**. Furthermore, as illustrated in FIG. **13**, a pair of front-side locked portions **27** are formed so as to protrude upward from the left and right front side ends of the cover wall **73**. The front-side locked portions **27** have the shape of rectangular frames, and are formed so as to be capable of deforming by bending in the left-right direction. As illustrated in FIG. **2**, the front-side locking portions **26** of the housing **12** can fit into front-side locked holes **82** formed in the center of the front-side locked portions **27**.
(63) The cover **15** includes an excessive-bend-preventing portion **87** that protrudes upward from the cover wall **73**. The excessive-bend-preventing portion **87** includes a pair of leg portions **88** that stand up from the left and right sides of the front end of the cover wall **73**, and a bridge portion **89** that bridges the upper ends of the leg portions **88**. The leg portions **88** have L-shapes in a lateral view, and are shaped so as to stand up from the cover wall **73** and then protrude toward the front. As illustrated in FIGS. **2** and **6**, the excessive-bend-preventing portion **87** covers the lower end side (free end side) of the lock arm **25** from the front side and both lateral sides in a state in which the cover **15** is attached to the housing **12**, and has the function of restricting the lock arm **25** from bending excessively.
(64) Connector Assembly Method and Fitting Structure of Connector and Mating Connector
(65) First, a method for assembling the connector **10** will be described. Note that, preferably, the work for attaching components to the housing **12** described below is carried out in a state in which the up-down-direction orientation of the housing **12** is up-side-down relative to that when the connector **10** is installed, so that the inside of the housing **12** is open upward.
(66) The terminal metal fittings **11** are connected to the terminal portions of the electric wires **200** in advance. In this state, the cylinder portions **17** of the terminal metal fittings **11** in the upper tier are inserted into the rear-side terminal insertion holes **33** (see FIG. **6**). The front sides of the extension portions **18** of the terminal metal fittings **11** in the upper tier are fitted into the recesses **34** of the fitting portion **24** and inserted into the terminal housing chambers **59** of the housing main body **23**. The terminal metal fittings **11** are positioned relative to the housing **12** by the protruding pieces **22** being fitted into the protruding-piece receiving portions **35** of the housing **12**. The

electric wires **200** connected to the terminal metal fittings **11** in the upper tier are drawn out toward the rear from the housing **12**.

(67) Subsequently, the upper inner member **13** is inserted into the housing **12**. In the process of inserting the upper inner member **13**, the upper inner lock portions **54** come into sliding contact with the upper guide portions **68**. The upper inner member **13** is retained by the housing **12** by the front end portion of the upper base portion **61** coming into contact with the upper-tier surface **29** and the upper inner lock portions **54** being fitted inside the upper inner lock receiving portions **69** (see FIG. **4**). The extension portions **18** of the terminal metal fittings **11** in the upper tier are sandwiched between the upper inner member **13** and the upper wall **37** of the housing main body **23**, and are restricted from falling out from the corresponding terminal housing chambers **59** (see FIG. **6**).

(68) Subsequently, the cylinder portions **17** of the terminal metal fittings **11** in the middle tier are inserted into the terminal insertion holes **33** between the front and rear sides, and the extension portions **18** of the terminal metal fittings **11** in the middle tier are inserted into the terminal housing chambers **59** of the upper inner member **13**. Next, the lower inner member **14** is inserted into the housing **12**. In the process of inserting the lower inner member **14**, the lower inner lock portions **47** come into sliding contact with the lower guide portions **76**. The lower inner member **14** is retained by the housing **12** by the front end portion of the lower base portion **65** coming into contact with the middle-tier surface **31** and the lower inner lock portions **47** being fitted inside the lower inner lock receiving portions **77** (see FIG. **5**). The extension portions **18** of the terminal metal fittings **11** in the middle tier are sandwiched between the upper inner member **13** and the lower inner member **14**, and are restricted from falling out from the corresponding terminal housing chambers **59** (see FIG. **6**).

(69) Similarly, the cylinder portions **17** of the terminal metal fittings **11** in the lower tier are inserted into the front-side terminal insertion holes **33**, and the extension portions **18** of the terminal metal fittings **11** in the lower tier are inserted into the terminal housing chambers **59** of the lower inner member **14**. The entire general terminal is inserted into a general terminal insertion hole **84** (see FIG. **9**) that is adjacent to the terminal insertion holes **33** inside the fitting portion **24**. Next, the cover **15** is inserted into the housing **12**. The rear part of the cover **15** is arranged inside the cover housing portion **38** of the housing main body **23**. The lateral walls **79** of the cover **15** face the inner surfaces of the locking walls **36**, and the locked portions **53** are fitted into and locked to the locking holes **52** in the locking portions **51** after the locking portions **51** are bent. At the same time, the front-side locking portions **26** are fitted into and locked to the front-side locked holes **82** in the front-side locked portions **27**. Thus, the cover **15** is retained by the housing **12**. The extension portions **18** of the terminal metal fittings **11** in the lower tier are sandwiched between the lower inner member **14** and the cover **15**, and are restricted from falling out from the corresponding terminal housing chambers **59** (see FIG. **6**). In such a manner, the assembly of the connector **10** is completed.

(70) Next, the connector **10** is fitted to the mating connector **8**. The mating connector **8** is fixed to the circuit board **100** above the roof panel **90**, and is arranged so that the guide walls **9** pass through the inside of the opening **91** in the roof panel **90** (see FIG. **2**) while the surface to be fit to the connector **10** is oriented downward.

(71) The fitting portion **24** of the connector **10** is inserted into the guide walls **9** of the mating connector **8** from below. Once the connector **10** and the mating connector **8** are fitted to one another properly, the lock arm **25** locks the mating connector **8**, and the connector **10** and the mating connector **8** are retained in the fitted state. The terminal metal fittings **11** and the general terminal are electrically connected to the corresponding mating terminal metal fittings **95** and tab inside the fitting portion **24**.

(72) In a state in which the connector **10** and the mating connector **8** are properly fitted to one another, the upper surface of the upper wall **37** of the housing main body **23** and the upper surfaces

of the eaves portions **62** are arranged close to and facing the lower surface of the roof panel **90** as illustrated in FIG. **1**.

(73) Incidentally, because the roof panel **90** is a metal plate made from iron or the like, frost (film of ice) or dew may be formed on the lower surface side of the roof panel **90** in cold areas, etc. Electrical reliability decreases if water generated from frost or dew moves to the inside of the housing **12** and comes into contact with the portions where the terminal metal fittings **11** and the electric wires **200** are connected to one another. This being the case, in the case of present embodiment 1, the pair of eaves portions **62** are formed so as to protrude toward the left and right outer sides from the upper end of the housing main body **23**. Furthermore, the hanging-down portions **86** are formed so as to protrude downward from the protruding ends of the eaves portions **62**. Thus, water is allowed to freely fall along the left and right outer lateral surfaces of the hanging-down portions **86** from the upper surfaces of the eaves portions **62** without coming into contact with the outer surfaces of the locking walls **36** of the housing **12**. Thus, the infiltration of water into the terminal housing chambers **59** from the outer surfaces of the locking walls **36** can be prevented, and electrical reliability between the terminal metal fittings **11** and the electric wires **200** can be ensured.

(74) Furthermore, because the lateral walls **79** of the cover **15** are arranged inside the locking walls **36**, and gaps between the locking walls **36** and the lateral walls **79** of the cover **15** are also located inside the cover housing portion **38**, the upper surface of which is closed off by the top surface portions **42**, the infiltration of water into these gaps can be avoided even if water comes into contact with the outer surfaces of the locking walls **36**. Accordingly, the infiltration of water into the terminal housing chambers **59** can be more reliably prevented according to present embodiment 1. Furthermore, the infiltration of water into the lower-tier extraction hole **48** and the upper-tier extraction hole **55** can also be avoided.

(75) As described above, according to present embodiment 1, the infiltration of water into the terminal housing chambers **59** can be prevented and waterproofness of the housing **12** can be ensured because the eaves portions **62** are formed so as to protrude laterally above the locking walls **36** locked to the cover **15**. In addition, waterproofness of the housing **12** can be improved because the cover **15** is arranged inside the cover housing portion **38** of the housing **12**, and thus a structure in which water does not readily infiltrate into gaps between the locking walls **36** of the housing **12** and the lateral walls **79** of the cover **15** is adopted.

(76) Furthermore, water can be prevented from remaining on the upper surface of the upper wall **37** of the housing main body **23** because the upper surfaces of the eaves portions **62** are continuous with the upper surface of the upper wall **37** without any level differences. Moreover, because the hanging-down portions **86** are formed so as to protrude downward from the eaves portions **62**, a structure is realized in which water falling from above is less likely to infiltrate into the terminal housing chambers **59**, and the waterproofness of the housing **12** can be improved to a further extent.

Other Embodiments of Present Disclosure

(77) Embodiment 1 disclosed herein is an example in every way, and shall be construed as being non-limiting.

(78) In the case of above-described embodiment 1, the housing is formed so that the upper inner member and the lower inner member can be separated from the housing. However, the housing may be formed integrally with portions corresponding to the upper inner member and the lower inner member so that such portions cannot be separated from the housing.

(79) In the case of above-described embodiment 1, the eaves portions are formed so as to protrude toward the left and right outer sides from the upper end of the housing main body. However, it is sufficient that the eaves portions be formed so as to protrude from the upper end of the housing main body in a direction that intersects the up-down direction (laterally), and may for example include portions that protrude toward the rear from the upper end of the housing main body.

Furthermore, an eaves portion may be formed so as to protrude laterally from the upper end of the housing over the entire circumference.

(80) In the case of above-described embodiment 1, the eaves portions are formed so as to be flat along the front-rear direction and the left-right direction. However, the eaves portions may be formed in the shape of curved surfaces or slopes, and may for example be shaped to urge water to fall by inclining the outer surfaces of the left and right outer end portions downward. The hanging-down portions may be inclined as well.

(81) In the case of above-described embodiment 1, the housing includes the hanging-down portions protruding downward from the eaves portions. However, the hanging-down portions may be omitted from the housing depending on situation.

(82) In the case of above-described embodiment 1, a connector that is arranged below a roof panel was described as an example. However, the present disclosure is widely applicable to connectors for use in environments in which water can fall down onto the connectors from above, without limitation to above-described embodiment 1.

(83) From the foregoing, it will be appreciated that various exemplary embodiments of the present disclosure have been described herein for purposes of illustration, and that various modifications may be made without departing from the scope and spirit of the present disclosure. Accordingly, the various exemplary embodiments disclosed herein are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. A connector comprising: a housing main body which includes a terminal housing chamber and an upper surface of which is closed off; a terminal metal fitting that is housed inside the terminal housing chamber; a cover that is locked to the housing main body and arranged below the terminal housing chamber; and an eaves portion that protrudes laterally from an upper end of the housing main body, wherein the housing main body includes a pair of locking walls, a cover housing portion having an upper surface which is closed off is formed between the pair of locking walls in the housing main body, and the cover is arranged inside the cover housing portion, and includes lateral walls that face inner surfaces of the locking walls and that are locked to the locking walls.
2. The connector according to claim 1, wherein the connector is a connector that is provided to a roof panel of a vehicle, and the eaves portion is arranged along a lower surface of the roof panel.
3. The connector according to claim 1, wherein the eaves portion is shaped so as to protrude laterally further than a locked portion where the housing main body and the cover are locked to one another does, above the locked portion.
4. The connector according to claim 3, wherein an upper surface of the eaves portion is continuous with the upper surface of the housing main body without any level differences.
5. The connector according to claim 1 further comprising: a hanging-down portion that protrudes downward from a laterally-protruding end side of the eaves portion.
6. The connector according to claim 5, wherein an upper portion of a space between the hanging-down portion and one of the locking walls is closed off by the eaves portion.
7. A connector comprising: a housing main body which includes a terminal housing chamber and an upper surface of which is closed off; a terminal metal fitting that is housed inside the terminal housing chamber; a cover that is locked to the housing main body and arranged below the terminal housing chamber; and an eaves portion that protrudes laterally from an upper end of the housing main body, wherein the connector is a connector that is provided to a roof panel of a vehicle, and the eaves portion is arranged along a lower surface of the roof panel.